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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 25: TOD implementation

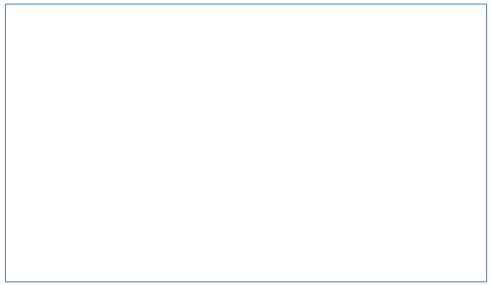
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TOD implementation:

- Features of inclusive & resilient TOD
- Barriers in TOD implementation
- Scale of TOD
- Stakeholder
- Readiness assessment
- Transit selection
- Corridor selection



Inclusive and resilient TOD 1/2

- Align human/economic densities, mass transit capacity and network characteristics for greater accessibility
- Create compact regions with short commutes
- Ensure resilience of areas connected by mass transit
- Plan for mixed-income neighborhoods at corridor level



Image source: World Economic Forum

Inclusive and resilient TOD 2/2

- Create vibrant, people-centric public spaces around stations
- Develop neighborhoods that promote walking and biking
- Develop good quality, accessible, and integrated public transit
- Manage private vehicle demand (minimize car use)



Image source: tripsavvy.com

Inclusive infrastructure

Inclusive Infrastructure:

- Enhances positive outcomes in social inclusivity
- No individual, community or social group is left behind
- No one is prevented from benefiting from improved infrastructure

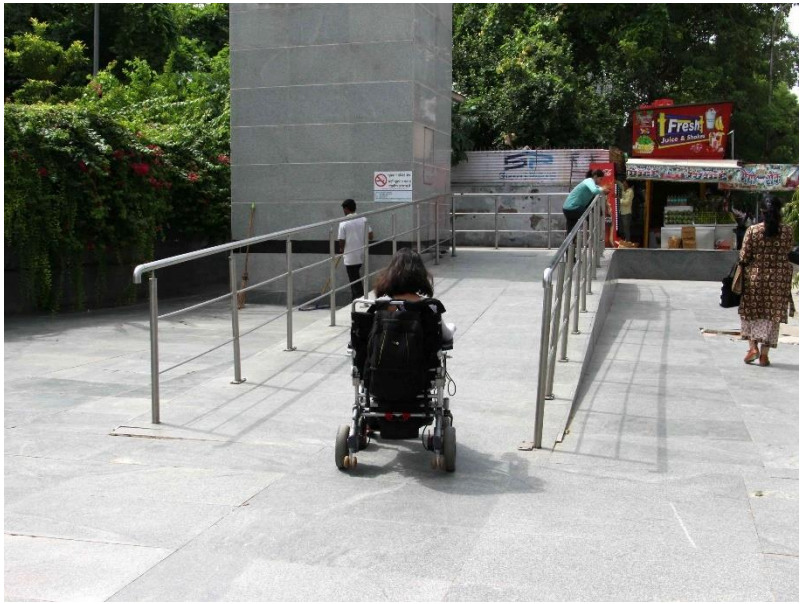


Image: Ramp for wheelchair makes transit accessible for PWDs.

Source: scroll.in

Example of Inclusive infrastructure: Delhi Metro

Delhi metro fare structure for 32 km+ trip is Rs 60 (avg. cost for 32 km= 1.87 INR/km), which is very low specially compared to cab/taxi.

This makes Delhi metro affordable even for poorest section of society and ensure everyone enjoys the benefits of public infrastructure.

While designing, Delhi metro, it is ensured that it is user friendly for Person with disability (PWD)/Divyangs through proper signage, guidance path, wheelchair mobility etc.



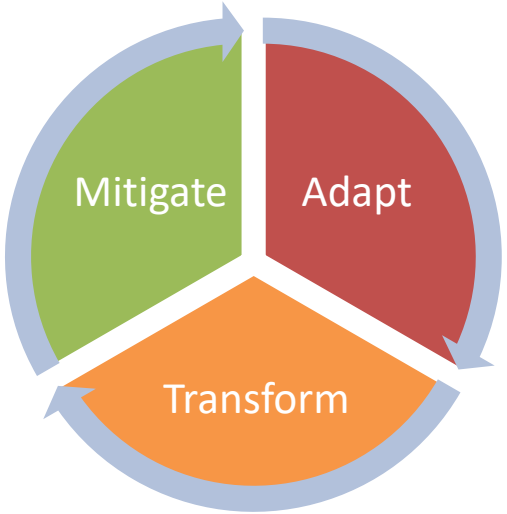
Source: urbanmobilityindia.in

Note: Fare prices of Delhi metro are flexible in nature.

Resilient infrastructure

Capacity of infrastructure & related systems to:

- Mitigate, adapt or positively respond to chronic and acute stresses (i.e., earthquake, tsunami, overcrowding etc.)
- Transforming system in ways that restore, maintain and even improve their essential functions



Resilience Cycle

Example of resilient infrastructure: Delhi

Delhi comes under one of most seismic active zone.
Delhi metro is resilient for earthquake but development along the route and structures above metro tunnels in old Delhi areas are not resilient to earthquake.



Image: High-rise apartment with a "soft storey" for parking on the ground floor making them more vulnerable to earthquakes

Source: United Nations Office for Disaster Risk Reduction, 2019
Image: latimes.com

Barriers in TOD implementation 1/3



- Lack of regional coordination at the metropolitan level

Example: Delhi Metro spread in three states, NCT-Delhi, Haryana & Uttar Pradesh. Initially lack coordination hurdled the implementation of metro system.

Example: Regional Coordination

- Initially Gurugram/rapid metro (in state of Haryana), was separately built and developed.
- Later operation was handed over to DMRC (Delhi Metro Rail corporation).
- It has brought more coordinated operation of rapid metro and single card system for users during ride.



Image source: deccanherald.com

Barriers in TOD implementation 2/3

- Sector silo behavior and practices at the city level
- Inadequate policies and regulations for strategically creating articulated densities
- Restrictive national regulations and administrative constraints

Example: Policies which don't permit any activity associated with construction or development of TOD in any region such as environmental policy or restrictions due to security etc.

Sector silo behavior : Hesitation in sharing data or information with other sectors (inter-departmental)

Articulated densities

- Sometimes density obtained from just dividing the population with area is deceptive in nature.

- Ex.- If a sub-urban district's overall population density is very less but most of the population living in a very dense land use, density must be assumed for that part of district not whole area.

- While planning TOD, it should be ensured that it is comfortable for users, travelling in transit shouldn't become tedious task.

- Land use planning should be done to properly manage the density.

Densification neither should be very less that it doesn't create enough ridership nor make it should over crowd the transit.

Example: Articulated densities- Overcrowding



Image: Crowd at Rajiv Chowk metro station

Source: Times of India

- Local market (Palika Bazar), Heritage structure & prominent market and as a recreational hub, Connaught place (Rajiv Chowk) has very high footfall.
- Also being the junction for Yellow and Blue line of DMRC, metro station becomes very crowded even after high frequency metro.
- This makes transit uncomfortable.

Barriers in TOD implementation 3/3

- Inconsistencies in the planning instruments and deficiencies in their implementation

- Inadequate policies, regulations, and supporting mechanisms for redeveloping built-up areas, particularly brownfields or distressed and blighted districts

- Neglected urban design at the neighborhood and street level

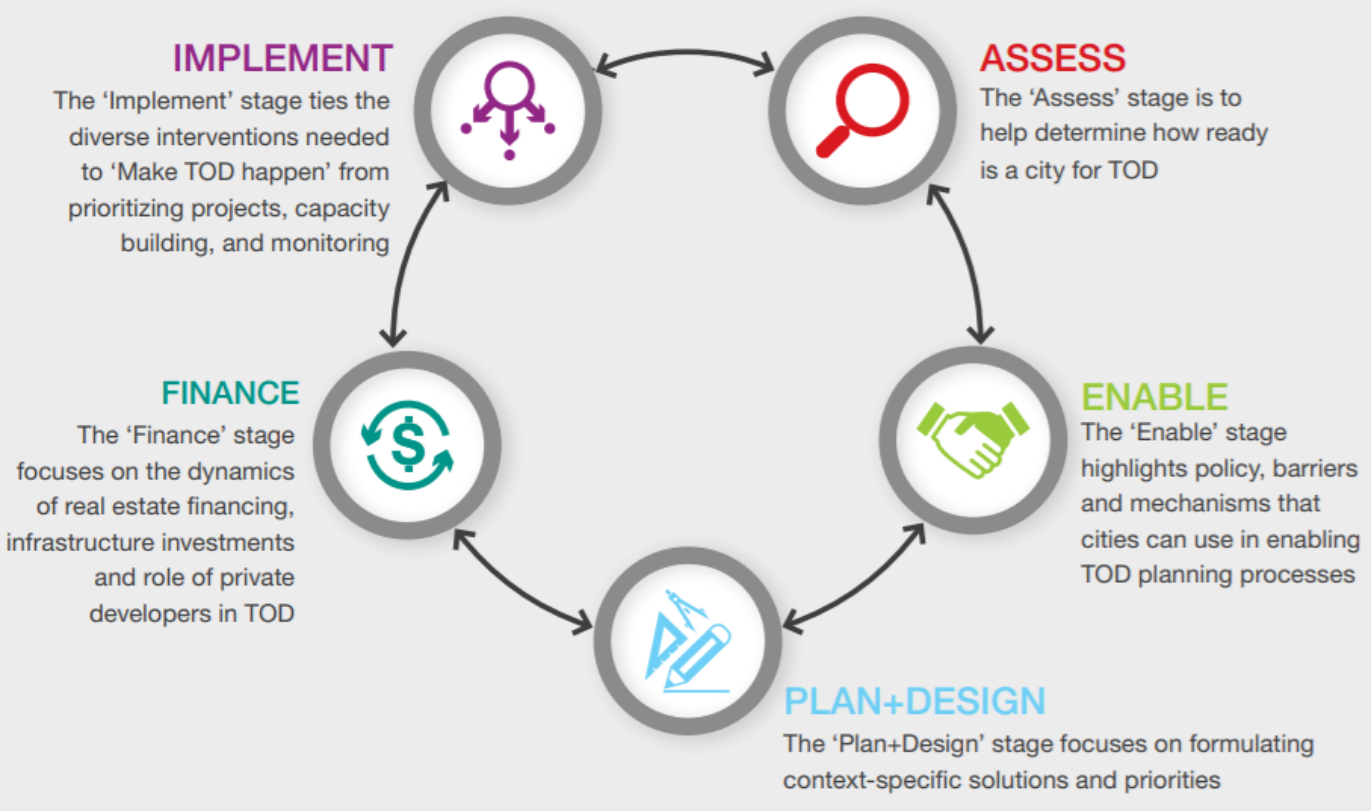
(Supporting road infrastructure is essential to assess the transit)

- Financial constraints

(Development of MRTS system is very cost intensive)

TOD aims at high rise-compact & mix use development but if high rise construction is not allowed, would implementation of TOD fully possible?
No, need to change policies & regulation accordingly.

TOD Framework



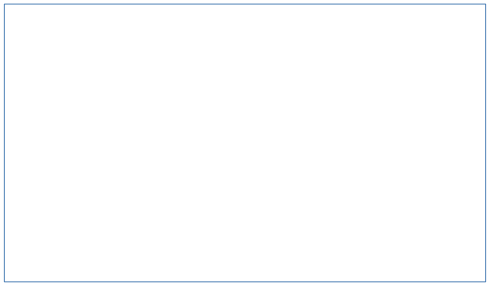
Scale of TOD

Area/location

- City region
- Corridor
- Station area
- Site level

Development context

- Greenfield
- Sub-urban
- Urban



Scale of TOD- City region

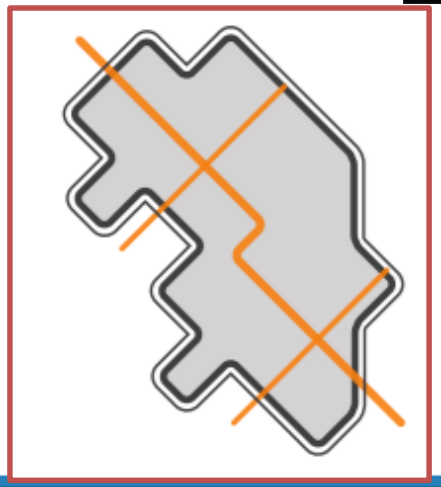
Administrative Boundaries/ Transit Systems

- Involves integration of land uses with transit system planning to support analysis and decision making related to citywide growth management.
- Provides a point of intervention for TOD as a policy in statutory documents (Master Plan/ Development Plan).



Image: TOD in Addis Ababa, Ethiopia

Source: nl4worldbank.org



Scale of TOD- Corridor

10-minute (800m-2km) walking/cycling distance on both sides of existing/planned transit corridor.

Ensures that development at one station complements development at other stations, resulting in a network of transit-oriented places.

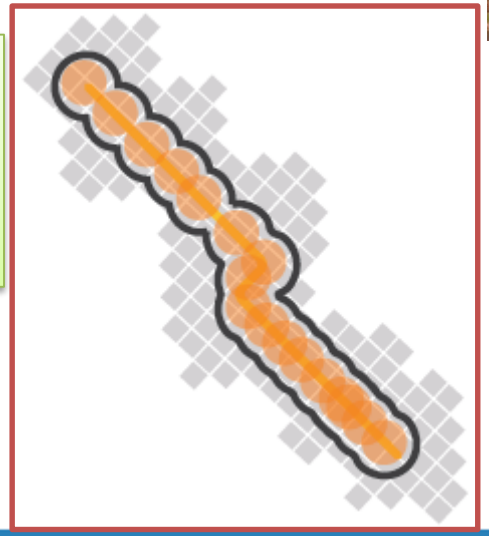




Image: Hubballi-Dharwad BRTS (twin city in Karnataka)
Source: deccanherald.com

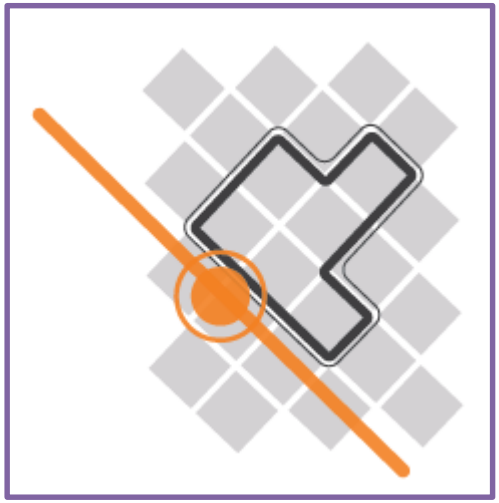
Scale of TOD- Station area



5–10-minute (400m-1km) walking distance from station facilities.
Ex.- TRX financial district, Kuala Lumpur, Malaysia

Focuses on areas surrounding transit stations within a 5-10minute walking distance focusing on land use, safe access to transit station for all users, transit station accessibility, multimodal integration and connectivity.

Scale of TOD- Site level



Individual parcel within 5–10-minute (800m-1km) walking distance from the station facility.

Ex.- Metro Mansion Station, Nanchang, China

Focuses on individual developments within a station area. Includes targets for net intensity and density for development, internal circulation, building design and parking.

TOD- Greenfield

Features of greenfield TOD

- Single ownership
- High percentage of government lands
- Lower land costs
- More financial resources appropriated
- Opportunity for constructing higher capacity infrastructure systems
- Minimal regulatory barriers



Example- Greenfield TOD: Naya Raipur

- Naya Raipur is satellite city of Raipur, capital of Chhattisgarh.
- A new city is planned with BRTS system.
- Land use planning is done along the transit following TOD principles.



TOD- Suburban



Features of Suburban TOD

- Higher percentage of sites available for transformation
- Opportunity to improve transit access to lower density neighborhoods
- Low land costs compare to urban

TOD- Urban/brownfield



Features of Suburban TOD

- Often located near major transportation corridors and established employment centers
- Active transportation modal share is higher, specifically in low- and middle-income areas
- Opportunity to improve transit access
- Opportunities for redeveloping aging building stock

Example: Urban TOD

- Delhi metro system is an example of Urban TOD.
- Delhi is already populated with dense land use.
- Later metro system was introduced for providing easy transit to residents.
- Appropriate zoning regulations were done to ensure compact land use development.



Image source: newindianexpress.com

Stakeholders in TOD planning

- Policy makers & political leadership
- Academia & researchers of related fields
- Transport planners
- Road (transport) safety experts
- Economic development stakeholders
- Community
- Urban planners

Stakeholder consultation



- Process where project proponent convince the policy makers, leadership, various stakeholders including civil society organization or groups to implement TOD.
- TOD projects are generally cost-incentive, huge changes in land use and affects a large section of society.
- Thus, it becomes important to bring all stakeholders on one page and put efforts uni-directionally to make a positive impact and transform the transits all along.

TOD readiness assessment

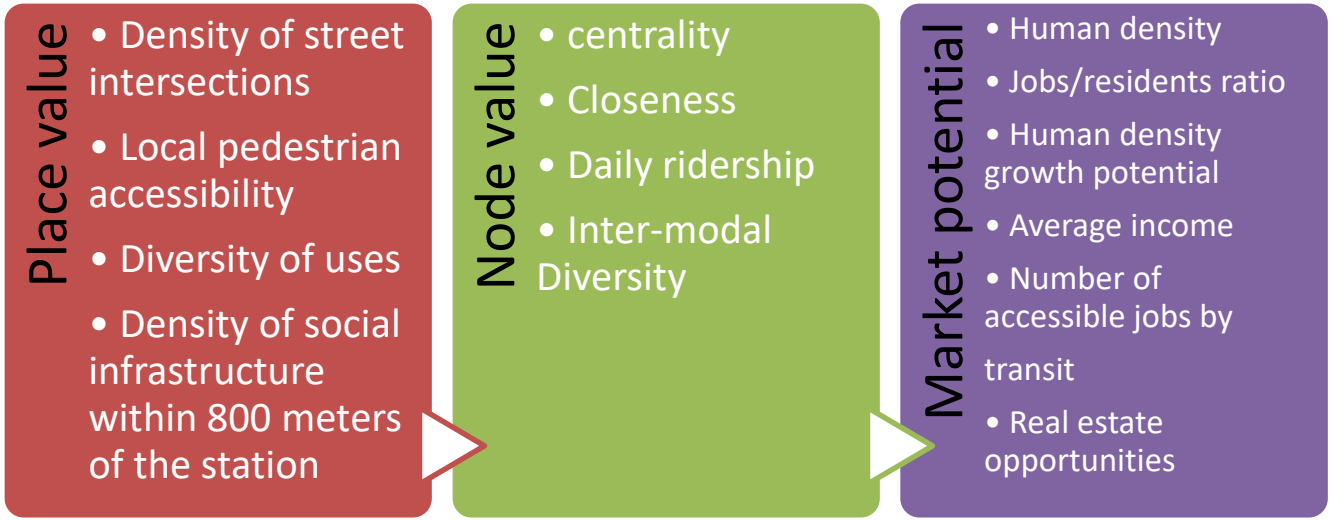
Assessment of any location/city/region based on following factors whether TOD can be implemented or not:

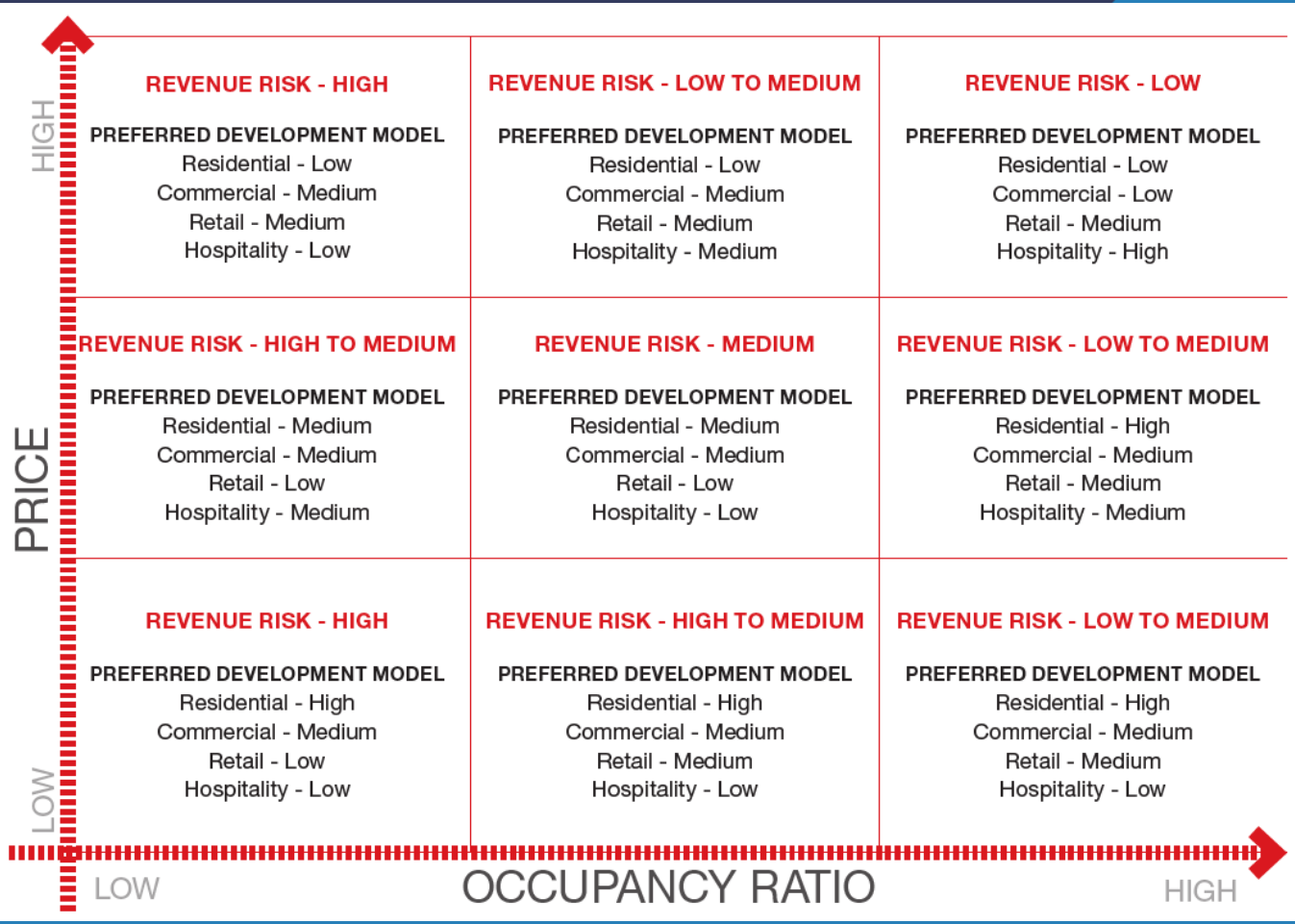
- Existing policy and regulatory framework
- Technical capacities available in-house
- Existing data availability to conduct detailed studies.

TOD readiness assessment: Data

- High-definition aerials/ satellite photography
- Census information
- Site Survey; Photos
- Local government transport data
- Secondary documents- applicable zoning codes; adopted master plans
- Google Street view or other similar applications

TOD detailed assessment: 3 Value Framework by World Bank





TOD -Revenue Risk Assessment

Selection of transit: Criteria

- Provides Adequate Capacity
- ROW Availability (Need vs. availability of dedicated corridors)
- Potential to Integrate Pedestrian Needs (Such as safe crossings)
- Potential to Improve Living Conditions in surrounding development
- Estimate of TAC per PKT (High Cost = Low Score)
- Ease of Implementation with respect to: Familiarity with technology

ROW- Right of way (way for transit)
TAC- Transport & accident cases
PKT- Passenger kilometer travelled

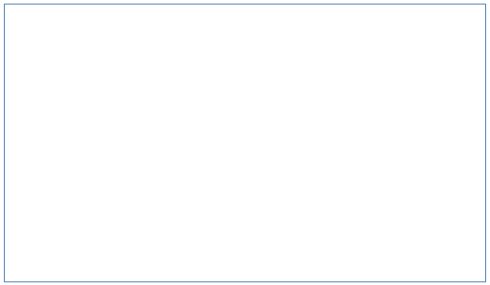
Transit alternative assessment

Criteria	Measure
People & job	Density of housing units and jobs identified through Census data and other surveys.
Destination & land use	Major trip generators within the city (weekdays & weekends) identified through destination mapping and land use maps.
Connection	Identification of existing and potential desired connections measured from travel data and people's perception
Existing policy	Review recommendations from existing Plans and Policy Documents and ensure they are still relevant & valid

Corridor selection criteria: City plan

Parameters for selection:

- Growth potential
- Economic development
- Mixed use
- Land value capture



Corridor selection criteria: Transportation Demand

Parameters for selection:

- Transit ridership
- Travel time
- Integration with existing network
- Reliability
- Safety
- Support of walk & NMT

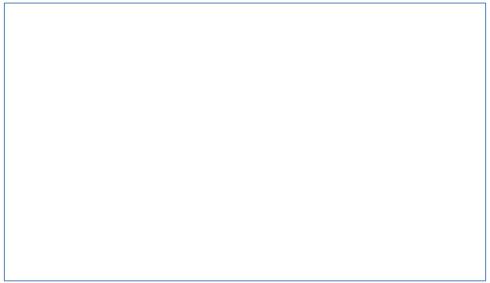
Corridor selection criteria: Implementation & operation

Parameters for selection:

- Ease of construction
- Financial viability
- Property impacts
- Environmental impacts
- Ability to implement

Conclusion

- There are many steps involved between idea to operation of TOD.
- Data driven assessment is required to plan and implement the TOD.
- TOD must be inclusive and resilient to ensure equal benefits to all and minimum risk for associated infrastructure.



References

- Ollivier, Gerald; Ghate, Ashish; Bankim, Kalra; Mehta, Prerna. 2021. Transit-Oriented Development Implementation Resources and Tools (2nd Edition). Global Platform for Sustainable Cities.

Thank You